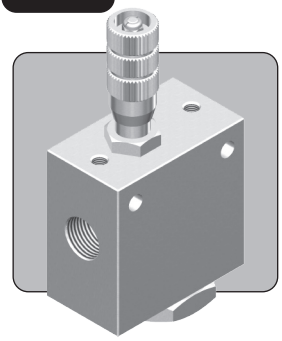
CMF50

METERING VALVE

1/2″

UP TO 414 BAR 6,000 PSI

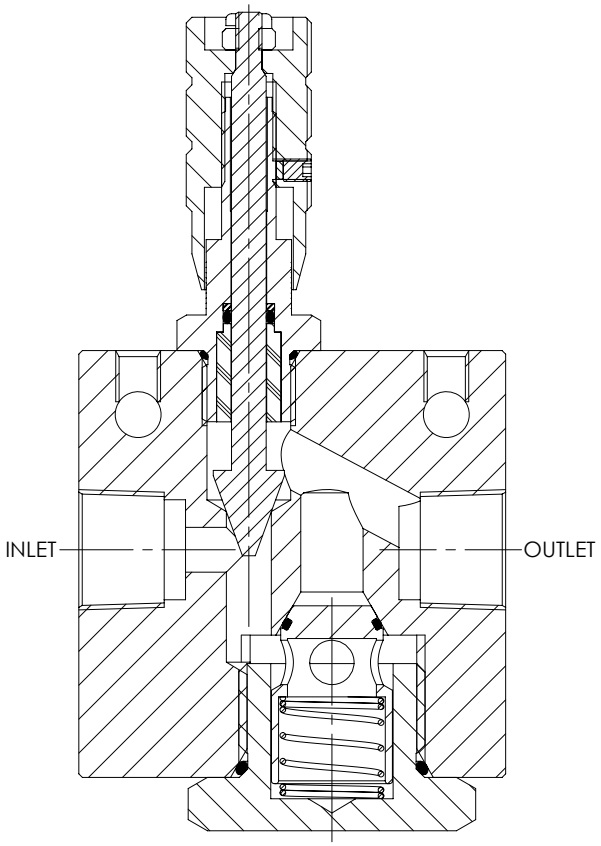


The CMF50 is a 1/2″ nominal bore metering valve with a reverse free-flow check function. It provides metered flow in one direction, and free flow in the opposite direction, ideal for uni-directional speed control of a cylinder or actuator.

The valve can be adjusted manually after installation, and the setting can be locked.

Note that this is a flow control valve and is not intended to provide tight shut-off when closed.

- Metering valve with reverse free-flow check feature
- Micrometer style adjustment with engraved graduations
- Non-rotating valve stem with lockable adjustment
- Ideal for uni-directional speed control
- Stainless steel construction suitable for liquid or gas
- Can be panel mounted
- Suitable for use with air, nitrogen, sweet natural gas, mineral oils, water glycols and plain water
- Suitable for many other media. Contact us for advice
- Various porting options available
- Various seal options available on request



Specifications

BASIC MODEL NUMBER	CMF50
SYMBOL	INLET (METERED) — OUTLET (FREE FLOW)
MAX WORKING PRESSURE (LIQUID)	414 bar (6,000 psi)
MAX WORKING PRESSURE (GAS)	207 bar (3,000 psi)
CV (FLOW CAPACITY)	Up to 1.4 See Typical Performance Graph
FLUID	Liquids and Gases See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2″
CHECK ELEMENT CRACKING PRESSURE	0.1 - 0.3 bar (2 - 5 psi)
WEIGHT	2.0 kg (4.4 lb)

Specifications may change without notice

Materials

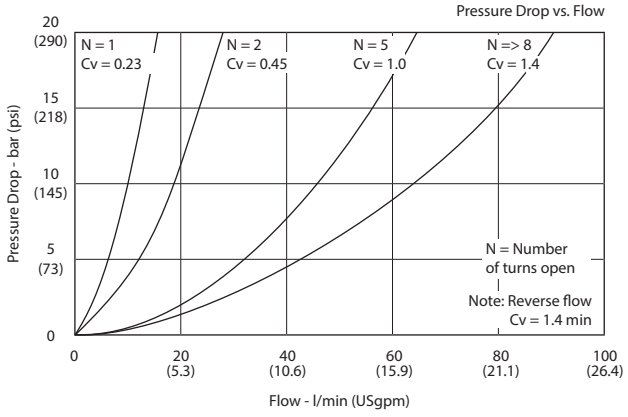
Externally Exposed Parts: 316, 302 and 17-4 PH stainless steel.

Internally Wetted Parts: 316, 302 and 17-4 PH stainless steel, and acetal.

The standard valve is designed for use with air, nitrogen, sweet natural gas, mineral oil, water glycols and plain water, but may be used with a wide variety of media compatible with the materials of construction. Other material options are available and for further advice, please contact us.

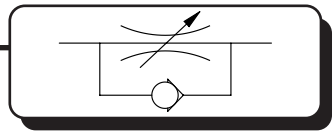
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

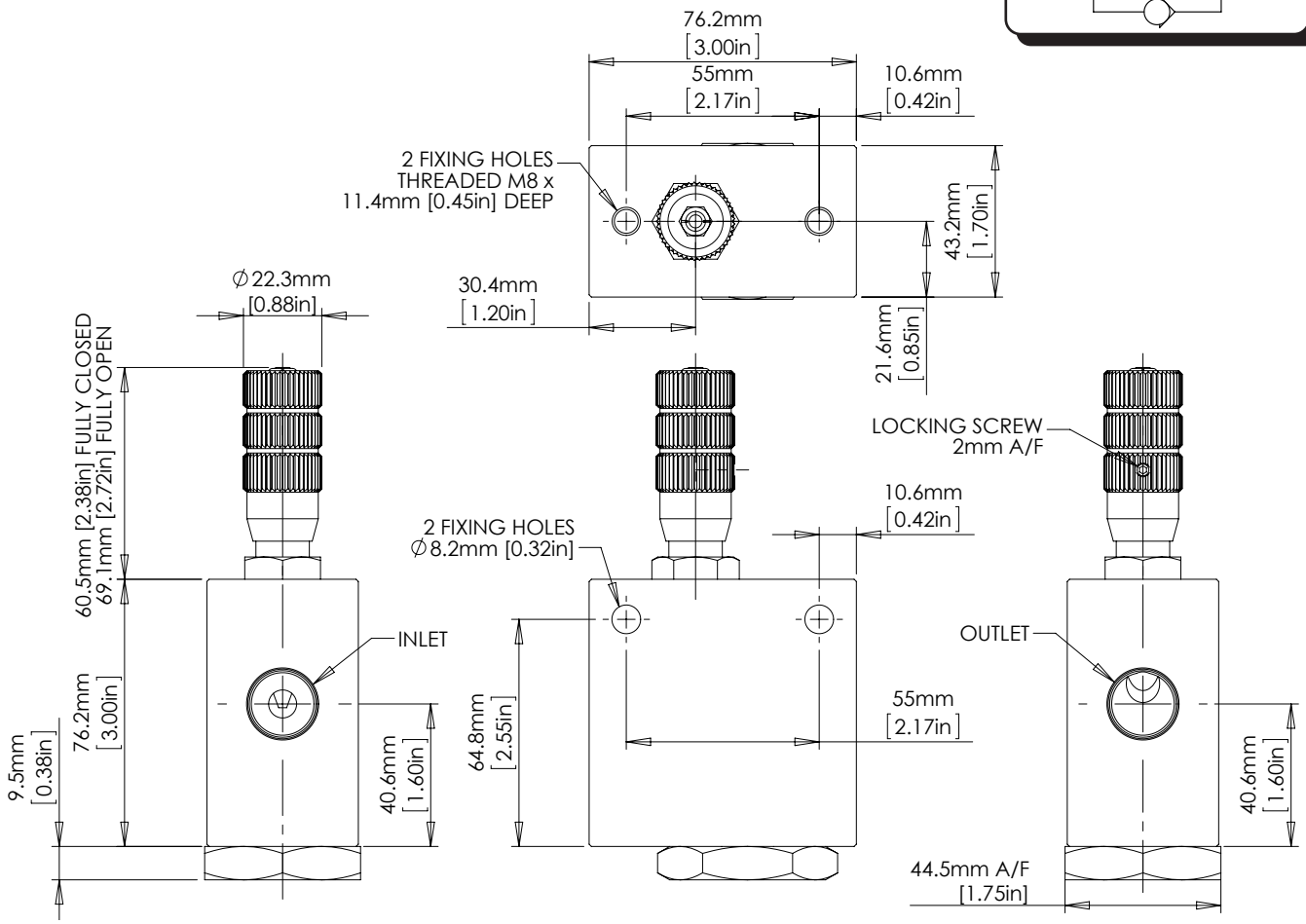
Typical Performance



Typical performance based on water

Installation Diagram





Product Selector

CMF50 P - NBR - 6K

VALVE TYPE	PORTING OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
CMF50	P 1/2" BSPP female N 1/2" NPT female	Leave blank for Viton® seals -10˚C to +120˚C NBR Nitrile seals -30˚C to +105˚C EP EPDM seals -45˚C to +120˚C KLZ Kalrez® seals -25˚C to +120˚C HNBR Hydrogenated Nitrile seals -20˚C to +120˚C Further seal options are available on request ⚠ Exposure to high and low fluid temperatures will influence handle temperature. Operator safety must be considered.	6K 414 bar (6,000 psi) Liquid 207 bar (3,000 psi) Gas

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK